Supplementary Material:
Evaluating the Resistance of ARADI Against Differential Fault Attack

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Overview

This document collects the material that the paper refers to the repository for: the figures of the round function, the key schedule and the fault propagation, the complete propagation tables, the table of column probabilities, and the extended experimental results. It is generated by `make_supplementary.py` from the cipher definition and from the experiment outputs in `results/` (seed 20251101, 10,000 keys per configuration). Section numbers in the text refer to the paper.

1 Figures of ARADI (Section II)

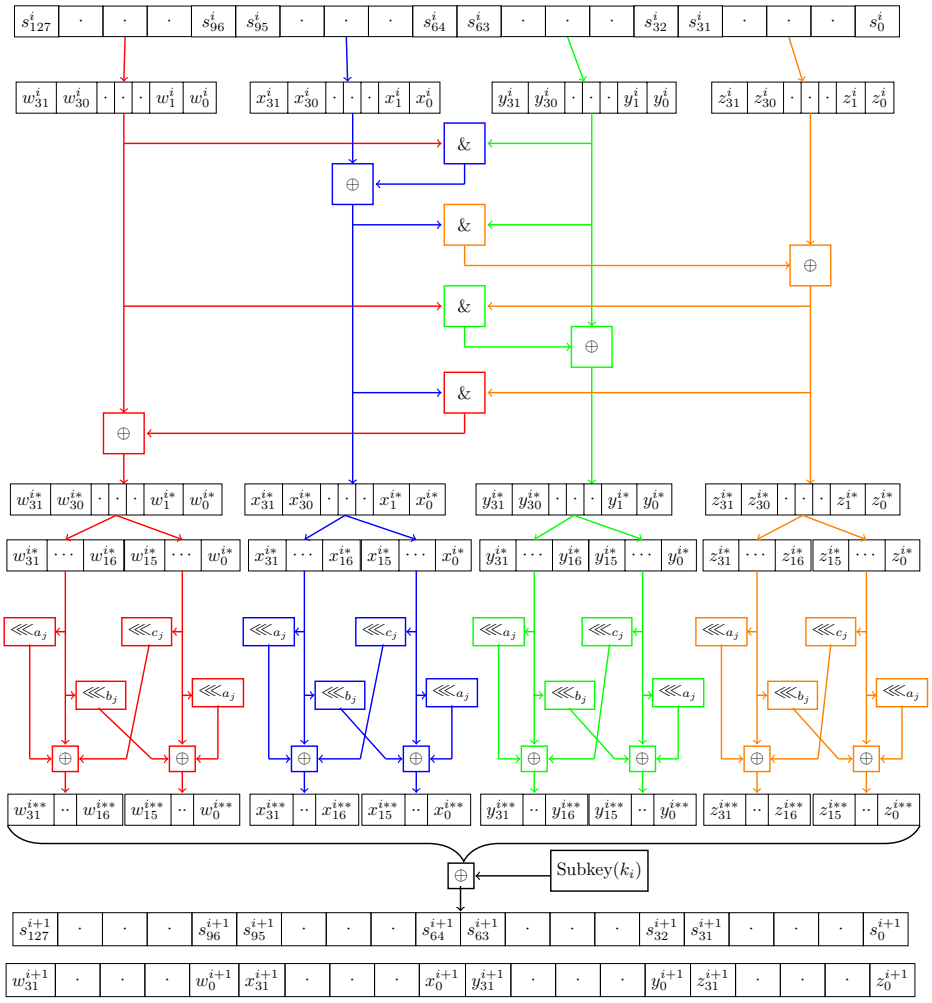


Figure 1: ARADI round function R_{i+1}, mapping S^i to S^{i+1}; S^{i*} denotes the state after the S-box layer.

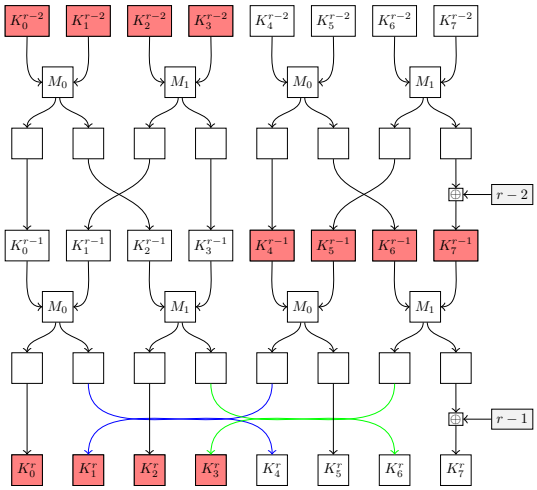


Figure 2: Two consecutive steps of the key schedule; red boxes are the words of the round keys.

Rotation constants of the linear layers and the S-box tables used by the code:

Table 1: Rotation constants of L_j (used in round i with $j = (i-1) \bmod 4$).

j	a_j	b_j	c_j
0	11	8	14
1	10	9	11
2	9	4	14
3	8	9	7

Table 2: The 4-bit S-box and its inverse (input $v = (w, x, y, z)$, w most significant).

v	0	1	2	3	4	5	6	7	8	9	A	B	C	D	E	F
$SB(v)$	0	1	2	3	4	D	F	6	8	B	5	E	C	7	A	9
$SB^{-1}(v)$	0	1	2	3	4	A	7	D	8	F	E	9	C	5	B	6

2 Fault propagation figures (Section III-B)

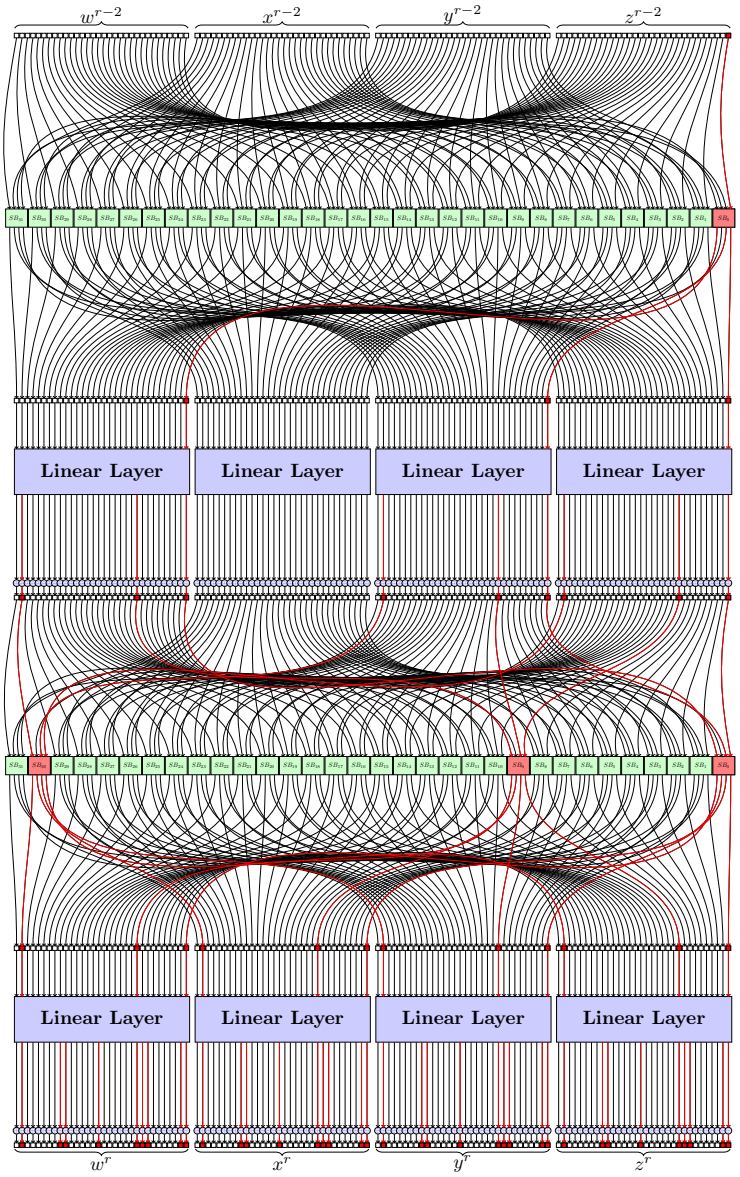


Figure 3: Propagation of a single-bit fault injected at $z^{14}[0]$ through the last two rounds (red: certainly active).

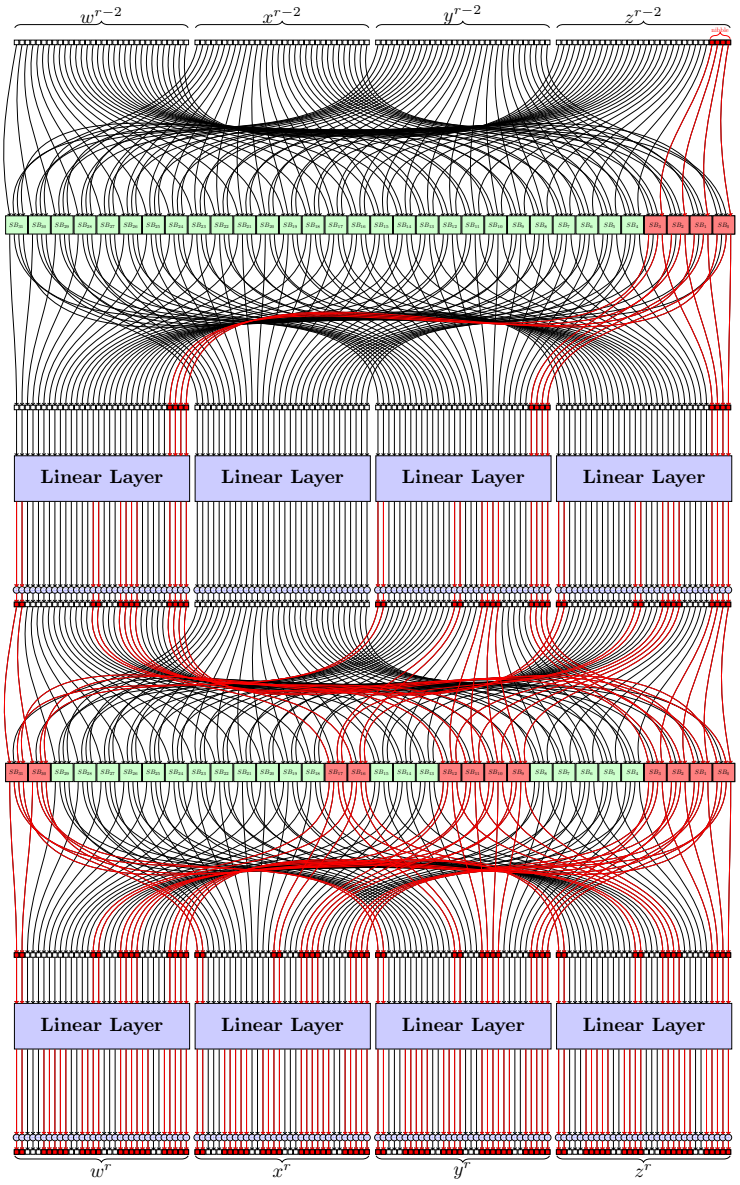


Figure 4: Propagation of the nibble fault `0x0000000f` injected in z^{14} through the last two rounds.

3 Propagation tables (Section III-B)

Conventions: a fault is injected at the start of round 15 into one word of S^{14}. *Certain* positions of the faulted word of S^{15} are $\mathrm{supp}(L_2(\Delta))$; the other words may be active at the same positions depending on the data, except that a fault in z never activates x^{15} (and a fault in x may activate all four words). The last column lists the positions of every ciphertext word that can be active, $\bigcup_{p\in\mathrm{supp}(L_2(\Delta))} \mathrm{supp}(L_3(e_p))$. The position lists are the same for a fault in z and a fault in x because the linear layer acts on every word in the same way; only the roles of the words differ.

Table 3: Bit-based propagation for a fault injected at $z^{14}[t]$.

t (faulty bit $z^{14}[t]$)	z^{15} certain; y^{15}, w^{15} possible	ciphertext words possible
0	0, 9, 30	0, 1, 7, 8, 9, 16, 22, 23, 30
1	1, 10, 31	1, 2, 8, 9, 10, 17, 23, 24, 31
2	2, 11, 16	2, 3, 9, 10, 11, 16, 18, 24, 25
3	3, 12, 17	3, 4, 10, 11, 12, 17, 19, 25, 26
4	4, 13, 18	4, 5, 11, 12, 13, 18, 20, 26, 27
5	5, 14, 19	5, 6, 12, 13, 14, 19, 21, 27, 28
6	6, 15, 20	6, 7, 13, 14, 15, 20, 22, 28, 29
7	0, 7, 21	0, 7, 8, 14, 15, 21, 23, 29, 30
8	1, 8, 22	0, 1, 8, 9, 15, 22, 24, 30, 31
9	2, 9, 23	0, 1, 2, 9, 10, 16, 23, 25, 31
10	3, 10, 24	1, 2, 3, 10, 11, 16, 17, 24, 26
11	4, 11, 25	2, 3, 4, 11, 12, 17, 18, 25, 27
12	5, 12, 26	3, 4, 5, 12, 13, 18, 19, 26, 28
13	6, 13, 27	4, 5, 6, 13, 14, 19, 20, 27, 29
14	7, 14, 28	5, 6, 7, 14, 15, 20, 21, 28, 30
15	8, 15, 29	0, 6, 7, 8, 15, 21, 22, 29, 31
16	4, 16, 25	2, 4, 9, 12, 16, 17, 24, 25, 27
17	5, 17, 26	3, 5, 10, 13, 17, 18, 25, 26, 28
18	6, 18, 27	4, 6, 11, 14, 18, 19, 26, 27, 29
19	7, 19, 28	5, 7, 12, 15, 19, 20, 27, 28, 30
20	8, 20, 29	0, 6, 8, 13, 20, 21, 28, 29, 31
21	9, 21, 30	1, 7, 9, 14, 16, 21, 22, 29, 30
22	10, 22, 31	2, 8, 10, 15, 17, 22, 23, 30, 31
23	11, 16, 23	0, 3, 9, 11, 16, 18, 23, 24, 31
24	12, 17, 24	1, 4, 10, 12, 16, 17, 19, 24, 25
25	13, 18, 25	2, 5, 11, 13, 17, 18, 20, 25, 26
26	14, 19, 26	3, 6, 12, 14, 18, 19, 21, 26, 27
27	15, 20, 27	4, 7, 13, 15, 19, 20, 22, 27, 28
28	0, 21, 28	0, 5, 8, 14, 20, 21, 23, 28, 29
29	1, 22, 29	1, 6, 9, 15, 21, 22, 24, 29, 30
30	2, 23, 30	0, 2, 7, 10, 22, 23, 25, 30, 31
31	3, 24, 31	1, 3, 8, 11, 16, 23, 24, 26, 31

Table 4: Bit-based propagation for a fault injected at $x^{14}[t]$.

t (faulty bit $x^{14}[t]$)	x^{15} certain; w^{15}, y^{15}, z^{15} possible	ciphertext words possible
0	0, 9, 30	0, 1, 7, 8, 9, 16, 22, 23, 30
1	1, 10, 31	1, 2, 8, 9, 10, 17, 23, 24, 31
2	2, 11, 16	2, 3, 9, 10, 11, 16, 18, 24, 25
3	3, 12, 17	3, 4, 10, 11, 12, 17, 19, 25, 26
4	4, 13, 18	4, 5, 11, 12, 13, 18, 20, 26, 27
5	5, 14, 19	5, 6, 12, 13, 14, 19, 21, 27, 28
6	6, 15, 20	6, 7, 13, 14, 15, 20, 22, 28, 29
7	0, 7, 21	0, 7, 8, 14, 15, 21, 23, 29, 30
8	1, 8, 22	0, 1, 8, 9, 15, 22, 24, 30, 31
9	2, 9, 23	0, 1, 2, 9, 10, 16, 23, 25, 31
10	3, 10, 24	1, 2, 3, 10, 11, 16, 17, 24, 26
11	4, 11, 25	2, 3, 4, 11, 12, 17, 18, 25, 27
12	5, 12, 26	3, 4, 5, 12, 13, 18, 19, 26, 28
13	6, 13, 27	4, 5, 6, 13, 14, 19, 20, 27, 29
14	7, 14, 28	5, 6, 7, 14, 15, 20, 21, 28, 30
15	8, 15, 29	0, 6, 7, 8, 15, 21, 22, 29, 31
16	4, 16, 25	2, 4, 9, 12, 16, 17, 24, 25, 27
17	5, 17, 26	3, 5, 10, 13, 17, 18, 25, 26, 28
18	6, 18, 27	4, 6, 11, 14, 18, 19, 26, 27, 29
19	7, 19, 28	5, 7, 12, 15, 19, 20, 27, 28, 30
20	8, 20, 29	0, 6, 8, 13, 20, 21, 28, 29, 31
21	9, 21, 30	1, 7, 9, 14, 16, 21, 22, 29, 30
22	10, 22, 31	2, 8, 10, 15, 17, 22, 23, 30, 31
23	11, 16, 23	0, 3, 9, 11, 16, 18, 23, 24, 31
24	12, 17, 24	1, 4, 10, 12, 16, 17, 19, 24, 25
25	13, 18, 25	2, 5, 11, 13, 17, 18, 20, 25, 26
26	14, 19, 26	3, 6, 12, 14, 18, 19, 21, 26, 27
27	15, 20, 27	4, 7, 13, 15, 19, 20, 22, 27, 28
28	0, 21, 28	0, 5, 8, 14, 20, 21, 23, 28, 29
29	1, 22, 29	1, 6, 9, 15, 21, 22, 24, 29, 30
30	2, 23, 30	0, 2, 7, 10, 22, 23, 25, 30, 31
31	3, 24, 31	1, 3, 8, 11, 16, 23, 24, 26, 31

Table 5: Nibble-based propagation for the 29 differences `0xf` $\ll k$ injected in z^{14}.

PND in z^{14}	z^{15} certain; y^{15}, w^{15} possible	ciphertext words possible
0x0000000f	0, 1, 2, 3, 9, 10, 11, 12, 16, 17, 30, 31	0, 1, 2, 3, 4, 7, 8, 9, 10, 11, 12, 16, 17, 18, 19, 22, 23, 24, 25, 26, 30, 31
0x0000001e	1, 2, 3, 4, 10, 11, 12, 13, 16, 17, 18, 31	1, 2, 3, 4, 5, 8, 9, 10, 11, 12, 13, 16, 17, 18, 19, 20, 23, 24, 25, 26, 27, 31
0x0000003c	2, 3, 4, 5, 11, 12, 13, 14, 16, 17, 18, 19	2, 3, 4, 5, 6, 9, 10, 11, 12, 13, 14, 16, 17, 18, 19, 20, 21, 24, 25, 26, 27, 28
0x00000078	3, 4, 5, 6, 12, 13, 14, 15, 17, 18, 19, 20	3, 4, 5, 6, 7, 10, 11, 12, 13, 14, 15, 17, 18, 19, 20, 21, 22, 25, 26, 27, 28, 29
0x000000f0	0, 4, 5, 6, 7, 13, 14, 15, 18, 19, 20, 21	0, 4, 5, 6, 7, 8, 11, 12, 13, 14, 15, 18, 19, 20, 21, 22, 23, 26, 27, 28, 29, 30
0x000001e0	0, 1, 5, 6, 7, 8, 14, 15, 19, 20, 21, 22	0, 1, 5, 6, 7, 8, 9, 12, 13, 14, 15, 19, 20, 21, 22, 23, 24, 27, 28, 29, 30, 31
0x000003c0	0, 1, 2, 6, 7, 8, 9, 15, 20, 21, 22, 23	0, 1, 2, 6, 7, 8, 9, 10, 13, 14, 15, 16, 20, 21, 22, 23, 24, 25, 28, 29, 30, 31
0x00000780	0, 1, 2, 3, 7, 8, 9, 10, 21, 22, 23, 24	0, 1, 2, 3, 7, 8, 9, 10, 11, 14, 15, 16, 17, 21, 22, 23, 24, 25, 26, 29, 30, 31
0x00000f00	1, 2, 3, 4, 8, 9, 10, 11, 22, 23, 24, 25	0, 1, 2, 3, 4, 8, 9, 10, 11, 12, 15, 16, 17, 18, 22, 23, 24, 25, 26, 27, 30, 31
0x00001e00	2, 3, 4, 5, 9, 10, 11, 12, 23, 24, 25, 26	0, 1, 2, 3, 4, 5, 9, 10, 11, 12, 13, 16, 17, 18, 19, 23, 24, 25, 26, 27, 28, 31
0x00003c00	3, 4, 5, 6, 10, 11, 12, 13, 24, 25, 26, 27	1, 2, 3, 4, 5, 6, 10, 11, 12, 13, 14, 16, 17, 18, 19, 20, 24, 25, 26, 27, 28, 29
0x00007800	4, 5, 6, 7, 11, 12, 13, 14, 25, 26, 27, 28	2, 3, 4, 5, 6, 7, 11, 12, 13, 14, 15, 17, 18, 19, 20, 21, 25, 26, 27, 28, 29, 30
0x0000f000	5, 6, 7, 8, 12, 13, 14, 15, 26, 27, 28, 29	0, 3, 4, 5, 6, 7, 8, 12, 13, 14, 15, 18, 19, 20, 21, 22, 26, 27, 28, 29, 30, 31
0x0001e000	4, 6, 7, 8, 13, 14, 15, 16, 25, 27, 28, 29	0, 2, 4, 5, 6, 7, 8, 9, 12, 13, 14, 15, 16, 17, 19, 20, 21, 22, 24, 25, 27, 28, 29, 30, 31
0x0003c000	4, 5, 7, 8, 14, 15, 16, 17, 25, 26, 28, 29	0, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 13, 14, 15, 16, 17, 18, 20, 21, 22, 24, 25, 26, 27, 28, 29, 30, 31
0x00078000	4, 5, 6, 8, 15, 16, 17, 18, 25, 26, 27, 29	0, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 21, 22, 24, 25, 26, 27, 28, 29, 31
0x000f0000	4, 5, 6, 7, 16, 17, 18, 19, 25, 26, 27, 28	2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 24, 25, 26, 27, 28, 29, 30
0x001e0000	5, 6, 7, 8, 17, 18, 19, 20, 26, 27, 28, 29	0, 3, 4, 5, 6, 7, 8, 10, 11, 12, 13, 14, 15, 17, 18, 19, 20, 21, 25, 26, 27, 28, 29, 30, 31
0x003c0000	6, 7, 8, 9, 18, 19, 20, 21, 27, 28, 29, 30	0, 1, 4, 5, 6, 7, 8, 9, 11, 12, 13, 14, 15, 16, 18, 19, 20, 21, 22, 26, 27, 28, 29, 30, 31
0x00780000	7, 8, 9, 10, 19, 20, 21, 22, 28, 29, 30, 31	0, 1, 2, 5, 6, 7, 8, 9, 10, 12, 13, 14, 15, 16, 17, 19, 20, 21, 22, 23, 27, 28, 29, 30, 31
0x00f00000	8, 9, 10, 11, 16, 20, 21, 22, 23, 29, 30, 31	0, 1, 2, 3, 6, 7, 8, 9, 10, 11, 13, 14, 15, 16, 17, 18, 20, 21, 22, 23, 24, 28, 29, 30, 31
0x01e00000	9, 10, 11, 12, 16, 17, 21, 22, 23, 24, 30, 31	0, 1, 2, 3, 4, 7, 8, 9, 10, 11, 12, 14, 15, 16, 17, 18, 19, 21, 22, 23, 24, 25, 29, 30, 31
0x03c00000	10, 11, 12, 13, 16, 17, 18, 22, 23, 24, 25, 31	0, 1, 2, 3, 4, 5, 8, 9, 10, 11, 12, 13, 15, 16, 17, 18, 19, 20, 22, 23, 24, 25, 26, 30, 31
0x07800000	11, 12, 13, 14, 16, 17, 18, 19, 23, 24, 25, 26	0, 1, 2, 3, 4, 5, 6, 9, 10, 11, 12, 13, 14, 16, 17, 18, 19, 20, 21, 23, 24, 25, 26, 27, 31
0x0f000000	12, 13, 14, 15, 17, 18, 19, 20, 24, 25, 26, 27	1, 2, 3, 4, 5, 6, 7, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 24, 25, 26, 27, 28
0x1e000000	0, 13, 14, 15, 18, 19, 20, 21, 25, 26, 27, 28	0, 2, 3, 4, 5, 6, 7, 8, 11, 12, 13, 14, 15, 17, 18, 19, 20, 21, 22, 23, 25, 26, 27, 28, 29
0x3c000000	0, 1, 14, 15, 19, 20, 21, 22, 26, 27, 28, 29	0, 1, 3, 4, 5, 6, 7, 8, 9, 12, 13, 14, 15, 18, 19, 20, 21, 22, 23, 24, 26, 27, 28, 29, 30
0x78000000	0, 1, 2, 15, 20, 21, 22, 23, 27, 28, 29, 30	0, 1, 2, 4, 5, 6, 7, 8, 9, 10, 13, 14, 15, 19, 20, 21, 22, 23, 24, 25, 27, 28, 29, 30, 31
0xf0000000	0, 1, 2, 3, 21, 22, 23, 24, 28, 29, 30, 31	0, 1, 2, 3, 5, 6, 7, 8, 9, 10, 11, 14, 15, 16, 20, 21, 22, 23, 24, 25, 26, 28, 29, 30, 31

Table 6: Nibble-based propagation for the 29 differences `0xf` $\ll k$ injected in x^{14}.

PND in x^{14}	x^{15} certain; w^{15}, y^{15}, z^{15} possible	ciphertext words possible
0x0000000f	0, 1, 2, 3, 9, 10, 11, 12, 16, 17, 30, 31	0, 1, 2, 3, 4, 7, 8, 9, 10, 11, 12, 16, 17, 18, 19, 22, 23, 24, 25, 26, 30, 31
0x0000001e	1, 2, 3, 4, 10, 11, 12, 13, 16, 17, 18, 31	1, 2, 3, 4, 5, 8, 9, 10, 11, 12, 13, 16, 17, 18, 19, 20, 23, 24, 25, 26, 27, 31
0x0000003c	2, 3, 4, 5, 11, 12, 13, 14, 16, 17, 18, 19	2, 3, 4, 5, 6, 9, 10, 11, 12, 13, 14, 16, 17, 18, 19, 20, 21, 24, 25, 26, 27, 28
0x00000078	3, 4, 5, 6, 12, 13, 14, 15, 17, 18, 19, 20	3, 4, 5, 6, 7, 10, 11, 12, 13, 14, 15, 17, 18, 19, 20, 21, 22, 25, 26, 27, 28, 29
0x000000f0	0, 4, 5, 6, 7, 13, 14, 15, 18, 19, 20, 21	0, 4, 5, 6, 7, 8, 11, 12, 13, 14, 15, 18, 19, 20, 21, 22, 23, 26, 27, 28, 29, 30
0x000001e0	0, 1, 5, 6, 7, 8, 14, 15, 19, 20, 21, 22	0, 1, 5, 6, 7, 8, 9, 12, 13, 14, 15, 19, 20, 21, 22, 23, 24, 27, 28, 29, 30, 31
0x000003c0	0, 1, 2, 6, 7, 8, 9, 15, 20, 21, 22, 23	0, 1, 2, 6, 7, 8, 9, 10, 13, 14, 15, 16, 20, 21, 22, 23, 24, 25, 28, 29, 30, 31
0x00000780	0, 1, 2, 3, 7, 8, 9, 10, 21, 22, 23, 24	0, 1, 2, 3, 7, 8, 9, 10, 11, 14, 15, 16, 17, 21, 22, 23, 24, 25, 26, 29, 30, 31
0x00000f00	1, 2, 3, 4, 8, 9, 10, 11, 22, 23, 24, 25	0, 1, 2, 3, 4, 8, 9, 10, 11, 12, 15, 16, 17, 18, 22, 23, 24, 25, 26, 27, 30, 31
0x00001e00	2, 3, 4, 5, 9, 10, 11, 12, 23, 24, 25, 26	0, 1, 2, 3, 4, 5, 9, 10, 11, 12, 13, 16, 17, 18, 19, 23, 24, 25, 26, 27, 28, 31
0x00003c00	3, 4, 5, 6, 10, 11, 12, 13, 24, 25, 26, 27	1, 2, 3, 4, 5, 6, 10, 11, 12, 13, 14, 16, 17, 18, 19, 20, 24, 25, 26, 27, 28, 29
0x00007800	4, 5, 6, 7, 11, 12, 13, 14, 25, 26, 27, 28	2, 3, 4, 5, 6, 7, 11, 12, 13, 14, 15, 17, 18, 19, 20, 21, 25, 26, 27, 28, 29, 30
0x0000f000	5, 6, 7, 8, 12, 13, 14, 15, 26, 27, 28, 29	0, 3, 4, 5, 6, 7, 8, 12, 13, 14, 15, 18, 19, 20, 21, 22, 26, 27, 28, 29, 30, 31
0x0001e000	4, 6, 7, 8, 13, 14, 15, 16, 25, 27, 28, 29	0, 2, 4, 5, 6, 7, 8, 9, 12, 13, 14, 15, 16, 17, 19, 20, 21, 22, 24, 25, 27, 28, 29, 30, 31
0x0003c000	4, 5, 7, 8, 14, 15, 16, 17, 25, 26, 28, 29	0, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 13, 14, 15, 16, 17, 18, 20, 21, 22, 24, 25, 26, 27, 28, 29, 30, 31
0x00078000	4, 5, 6, 8, 15, 16, 17, 18, 25, 26, 27, 29	0, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 21, 22, 24, 25, 26, 27, 28, 29, 31
0x000f0000	4, 5, 6, 7, 16, 17, 18, 19, 25, 26, 27, 28	2, 3, 4, 5, 6, 7, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 24, 25, 26, 27, 28, 29, 30
0x001e0000	5, 6, 7, 8, 17, 18, 19, 20, 26, 27, 28, 29	0, 3, 4, 5, 6, 7, 8, 10, 11, 12, 13, 14, 15, 17, 18, 19, 20, 21, 25, 26, 27, 28, 29, 30, 31
0x003c0000	6, 7, 8, 9, 18, 19, 20, 21, 27, 28, 29, 30	0, 1, 4, 5, 6, 7, 8, 9, 11, 12, 13, 14, 15, 16, 18, 19, 20, 21, 22, 26, 27, 28, 29, 30, 31
0x00780000	7, 8, 9, 10, 19, 20, 21, 22, 28, 29, 30, 31	0, 1, 2, 5, 6, 7, 8, 9, 10, 12, 13, 14, 15, 16, 17, 19, 20, 21, 22, 23, 27, 28, 29, 30, 31
0x00f00000	8, 9, 10, 11, 16, 20, 21, 22, 23, 29, 30, 31	0, 1, 2, 3, 6, 7, 8, 9, 10, 11, 13, 14, 15, 16, 17, 18, 20, 21, 22, 23, 24, 28, 29, 30, 31
0x01e00000	9, 10, 11, 12, 16, 17, 21, 22, 23, 24, 30, 31	0, 1, 2, 3, 4, 7, 8, 9, 10, 11, 12, 14, 15, 16, 17, 18, 19, 21, 22, 23, 24, 25, 29, 30, 31
0x03c00000	10, 11, 12, 13, 16, 17, 18, 22, 23, 24, 25, 31	0, 1, 2, 3, 4, 5, 8, 9, 10, 11, 12, 13, 15, 16, 17, 18, 19, 20, 22, 23, 24, 25, 26, 30, 31
0x07800000	11, 12, 13, 14, 16, 17, 18, 19, 23, 24, 25, 26	0, 1, 2, 3, 4, 5, 6, 9, 10, 11, 12, 13, 14, 16, 17, 18, 19, 20, 21, 23, 24, 25, 26, 27, 31
0x0f000000	12, 13, 14, 15, 17, 18, 19, 20, 24, 25, 26, 27	1, 2, 3, 4, 5, 6, 7, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 24, 25, 26, 27, 28
0x1e000000	0, 13, 14, 15, 18, 19, 20, 21, 25, 26, 27, 28	0, 2, 3, 4, 5, 6, 7, 8, 11, 12, 13, 14, 15, 17, 18, 19, 20, 21, 22, 23, 25, 26, 27, 28, 29
0x3c000000	0, 1, 14, 15, 19, 20, 21, 22, 26, 27, 28, 29	0, 1, 3, 4, 5, 6, 7, 8, 9, 12, 13, 14, 15, 18, 19, 20, 21, 22, 23, 24, 26, 27, 28, 29, 30
0x78000000	0, 1, 2, 15, 20, 21, 22, 23, 27, 28, 29, 30	0, 1, 2, 4, 5, 6, 7, 8, 9, 10, 13, 14, 15, 19, 20, 21, 22, 23, 24, 25, 27, 28, 29, 30, 31
0xf0000000	0, 1, 2, 3, 21, 22, 23, 24, 28, 29, 30, 31	0, 1, 2, 3, 5, 6, 7, 8, 9, 10, 11, 14, 15, 16, 20, 21, 22, 23, 24, 25, 26, 28, 29, 30, 31

The activity pattern of each of the 239 PNDs under L_2, with the cancellation check, is in `tables/all_pnd_patterns_L2.csv`.

Table 7: Propagation of the 239 PNDs through each linear layer: number of cancellation-free PNDs, whether the certain patterns are pairwise distinct, and number of activity patterns shared by two PNDs (always 0).

layer	cancellation-free PNDs	with cancellation	certain patterns distinct	ambiguous patterns	PNDs with cancellation
L_0	227	12	yes	0	0x12000, 0x16000, 0x1a000, 0x1e000, 0x24000, 0x2c000, 0x34000, 0x3c000, 0x48000, 0x58000, 0x68000, 0x78000
L_1	231	8	yes	0	0x18000, 0x1a000, 0x1c000, 0x1e000, 0x38000, 0x3c000, 0x58000, 0x78000
L_2	239	0	yes	0	–
L_3	239	0	yes	0	–

Table 8: Differences confined to m adjacent bits, propagated through L_2: identification is unique up to $m = 7$; byte faults ($m = 8$) have 5 ambiguous patterns.

window width m	differences	cancellation-free	certain patterns distinct	ambiguous patterns	differences involved
1	32	32	yes	0	0
2	63	63	yes	0	0
3	123	123	yes	0	0
4	239	239	yes	0	0
5	463	463	yes	0	0
6	895	815	yes	0	0
7	1727	1519	yes	0	0
8	3327	1667	yes	5	10

4 Probability that a PND activates a column (Section III-F)

Table 9: Probability that one random PND makes column p of z^{15} active: closed form $\frac{8}{15} \cdot \frac{1}{29} \sum_{t \in T_p} n(t)$ and simulation with 200,000 PNDs (max. deviation 0.0019); with aligned nibbles the probability is $3/15 = 0.2$ for every column.

p	T_p (source bits)	$\sum n(t)$	theory (sliding window)	simulation (sliding)	simulation (aligned nibbles)
0	0, 7, 28	9	0.1655	0.1642	0.2003
1	1, 8, 29	9	0.1655	0.1656	0.1989
2	2, 9, 30	9	0.1655	0.1657	0.2003
3	3, 10, 31	9	0.1655	0.1646	0.2001
4	4, 11, 16	12	0.2207	0.2209	0.1995
5	5, 12, 17	12	0.2207	0.2190	0.2009
6	6, 13, 18	12	0.2207	0.2196	0.2005
7	7, 14, 19	12	0.2207	0.2192	0.2014
8	8, 15, 20	12	0.2207	0.2205	0.1998
9	0, 9, 21	9	0.1655	0.1667	0.1991
10	1, 10, 22	10	0.1839	0.1829	0.2005
11	2, 11, 23	11	0.2023	0.2036	0.2004
12	3, 12, 24	12	0.2207	0.2207	0.1993
13	4, 13, 25	12	0.2207	0.2205	0.1990
14	5, 14, 26	12	0.2207	0.2211	0.2005
15	6, 15, 27	12	0.2207	0.2207	0.2001
16	2, 16, 23	11	0.2023	0.2030	0.1999
17	3, 17, 24	12	0.2207	0.2203	0.2006
18	4, 18, 25	12	0.2207	0.2192	0.1990
19	5, 19, 26	12	0.2207	0.2201	0.2001
20	6, 20, 27	12	0.2207	0.2188	0.2001
21	7, 21, 28	12	0.2207	0.2201	0.2000
22	8, 22, 29	11	0.2023	0.2023	0.1999
23	9, 23, 30	10	0.1839	0.1838	0.1988
24	10, 24, 31	9	0.1655	0.1643	0.2003
25	11, 16, 25	12	0.2207	0.2209	0.1995
26	12, 17, 26	12	0.2207	0.2212	0.1994
27	13, 18, 27	12	0.2207	0.2206	0.2000
28	14, 19, 28	12	0.2207	0.2213	0.2012
29	15, 20, 29	11	0.2023	0.2026	0.1998
30	0, 21, 30	7	0.1287	0.1295	0.1992
31	1, 22, 31	7	0.1287	0.1287	0.2002

5 Filtering power of one condition (Section III-C)

Table 10: Probability that a wrong key value $\kappa_p \oplus \varepsilon$ survives one condition, averaged over the S-box input and the admissible input differences, for each condition type.

wrong offset ε $(wxyz)$	$\Delta z = 1$ only	$\mathcal{D}(1)$: $\Delta x = 0, \Delta z = 1$	$\mathcal{D}(0)$: $\Delta x = 0, \Delta z = 0$	word fault: $\Delta x = 0$
0001	0.625	0.281	0.208	0.429
0010	0.500	0.375	0.167	0.429
0011	0.625	0.406	0.375	0.571
0100	0.500	0.281	0.208	0.571
0101	0.500	0.281	0.208	0.429
0110	0.500	0.281	0.208	0.429
0111	0.500	0.281	0.208	0.571
1000	0.625	0.281	0.208	0.429
1001	0.625	0.375	0.167	0.429
1010	0.625	0.281	0.208	0.429
1011	0.625	0.500	0.333	0.571
1100	0.500	0.375	0.167	0.571
1101	0.500	0.375	0.167	0.429
1110	0.500	0.375	0.167	0.429
1111	0.500	0.250	0.333	0.571
mean	0.550	0.333	0.222	0.486

6 Number of faults until the round key is unique (Section III-F)

Table 11: Number of injections until all 32 columns of the equivalent key are unique (10,000 keys per row; fresh plaintext per injection).

configuration	mean	std	min	median	90%	95%	99%	max	not unique within 400
nibble, $\Delta z = 1$ only, round 15	97.6	27.0	45	92	132	148	191	297	0
nibble, conditions $\mathcal{D}$, round 15	50.9	14.3	23	48	69	77	97	145	0
aligned nibbles, conditions $\mathcal{D}$, round 15	46.7	11.4	20	45	61	68	82	122	0
single bit, round 15	102.9	24.4	47	99	135	148	177	263	0
byte window, round 15	29.0	8.5	12	28	40	45	57	96	0
whole word, round 15	15.1	3.1	8	15	19	21	25	35	0
nibble, conditions $\mathcal{D}$, round 14	49.4	12.6	21	47	66	73	88	139	0
nibble, conditions $\mathcal{D}$, round 13	51.1	14.1	22	49	69	78	96	160	0
nibble, $\Delta z = 1$ only, round 14	95.5	24.1	43	91	127	139	173	307	0
nibble, $\Delta z = 1$ only, round 13	98.0	26.1	45	93	131	146	184	307	0

Table 12: Residual key space of the last-round key after f PNDs, $\log_2$ of the number of candidates (10,000 keys): left block condition $\Delta z = 1$ only, right block conditions $\mathcal{D}$.

f	mean ($\Delta z = 1$)	95%	max	unique	mean ($\mathcal{D}$)	95%	max	unique
12	74.79	86.2	97.8	0.0%	41.04	55.7	72.9	0.0%
18	54.60	66.6	79.8	0.0%	21.81	34.1	54.5	0.0%
24	38.96	50.6	69.7	0.0%	11.44	20.8	36.5	0.1%
30	27.48	37.9	52.3	0.0%	5.97	12.9	24.2	2.2%
36	19.19	28.3	42.8	0.0%	3.17	8.2	20.0	11.8%
42	13.35	21.2	32.4	0.0%	1.69	5.3	19.5	30.2%
48	9.30	16.0	28.1	0.1%	0.91	3.6	14.9	50.8%
54	6.48	12.0	24.8	0.5%	0.51	2.3	12.2	67.2%
60	4.55	9.0	23.1	2.1%	0.28	1.6	11.2	79.8%
72	2.28	5.6	14.3	13.8%	0.09	1.0	4.0	92.5%
84	1.17	3.6	9.6	35.3%	0.03	0.0	3.6	97.2%
96	0.62	2.0	6.6	57.0%	0.01	0.0	2.6	99.0%
108	0.34	2.0	5.0	73.1%	0.01	0.0	2.0	99.5%
120	0.19	1.0	4.3	83.8%	0.00	0.0	2.0	99.8%

Table 13: Candidates for the 16 key bits of one group as a function of the number of conditions it received (2,000 keys, all groups).

conditions	$\log_2$ cand. ($\Delta z = 1$)	95%	unique	$\log_2$ cand. ($\mathcal{D}$)	95%	unique
0	16.00	16.0	0.0%	16.00	16.0	0.0%
1	15.23	15.6	0.0%	14.42	15.3	0.0%
2	14.47	15.2	0.0%	12.88	14.3	0.0%
3	13.72	14.8	0.0%	11.45	13.3	0.0%
4	12.98	14.2	0.0%	10.11	12.3	0.0%
5	12.26	13.8	0.0%	8.92	11.3	0.0%
6	11.58	13.2	0.0%	7.82	10.3	0.0%
7	10.92	12.6	0.0%	6.87	9.6	0.0%
8	10.24	12.2	0.0%	5.98	8.6	0.0%
9	9.62	11.6	0.0%	5.22	8.0	0.2%
10	9.02	11.2	0.0%	4.53	7.3	0.7%
11	8.45	10.6	0.0%	3.92	6.6	1.7%
12	7.90	10.2	0.0%	3.42	6.2	3.4%
13	7.38	9.6	0.0%	2.97	5.6	5.6%
14	6.87	9.2	0.0%	2.55	5.2	9.7%
15	6.40	8.8	0.0%	2.19	4.9	13.8%
16	5.95	8.3	0.0%	1.91	4.3	18.5%
17	5.55	7.9	0.0%	1.65	4.0	23.2%
18	5.14	7.6	0.1%	1.42	3.9	29.6%
19	4.77	7.2	0.1%	1.23	3.3	35.1%
20	4.41	6.9	0.3%	1.06	3.0	40.8%
21	4.10	6.6	0.5%	0.93	3.0	46.2%
22	3.78	6.2	0.9%	0.79	3.0	51.5%
23	3.51	6.0	1.4%	0.67	2.6	56.7%
24	3.25	5.6	2.0%	0.58	2.6	61.8%
25	3.00	5.3	2.8%	0.50	2.0	66.1%
26	2.78	5.0	3.7%	0.43	2.0	69.6%
27	2.57	5.0	5.3%	0.38	2.0	72.9%
28	2.37	4.6	6.8%	0.32	1.6	76.5%
29	2.16	4.3	9.0%	0.28	1.6	78.8%
30	2.00	4.0	11.4%	0.24	1.6	81.7%
31	1.84	4.0	13.7%	0.21	1.0	83.9%
32	1.71	4.0	16.2%	0.18	1.0	85.8%
33	1.58	3.6	18.4%	0.16	1.0	87.6%
34	1.46	3.6	21.6%	0.14	1.0	88.9%
35	1.34	3.2	24.8%	0.13	1.0	89.9%
36	1.22	3.0	28.2%	0.10	1.0	91.9%
37	1.14	3.0	30.8%	0.09	1.0	92.2%
38	1.06	3.0	34.1%	0.08	1.0	93.2%
39	0.96	3.0	38.4%	0.07	1.0	94.2%
40	0.90	2.6	40.4%	0.06	0.0	95.0%
41	0.82	2.6	44.0%	0.05	0.0	95.7%
42	0.76	2.3	46.9%	0.04	0.0	96.3%
43	0.70	2.0	50.0%	0.04	0.0	96.7%
44	0.65	2.0	52.6%	0.03	0.0	97.0%
45	0.60	2.0	54.9%	0.03	0.0	97.5%
46	0.55	2.0	58.0%	0.03	0.0	97.8%
47	0.52	2.0	60.2%	0.02	0.0	98.1%
48	0.47	2.0	63.1%	0.02	0.0	98.3%

7 Same plaintext for every injection (Section III-F)

Table 14: Residual key space ($\log_2$) after f PNDs when every injection uses the same plaintext (1,000 keys): the key space stops shrinking, so distinct plaintexts are necessary.

f	mean ($\Delta z = 1$)	min	max	mean ($\mathcal{D}$)	min	max
12	91.9	67.2	108.4	57.9	29.6	87.8
24	79.1	58.7	97.3	37.2	14.3	66.0
36	73.8	57.9	92.6	29.6	9.0	56.0
48	71.7	55.3	89.8	26.8	9.0	52.6
72	70.2	51.8	89.0	24.8	9.0	46.7
96	69.7	51.8	86.2	24.2	9.0	44.1
120	69.5	51.8	86.2	24.0	9.0	44.1

8 End-to-end master-key recovery (Section III-F)

Table 15: End-to-end recovery of the master key (1,000 keys per row). The one failure in the 36+36+36 row had $2^{22.5}$ candidates for k_{16}, above the enumeration limit of 2^{18}.

conditions	PNDs per campaign	success	faults mean	max	$\log_2 \lvert\mathcal{K}_{16}\rvert$ mean	max	tested mean	max	pairs to reject mean	max	time ms mean	max
conditions $\mathcal{D}$	36+36+36	99.9%	108	108	3.09	22.5	142	65536	1.8	14	15	2965
conditions $\mathcal{D}$	48+48+48	100.0%	144	144	0.89	13.5	16	11520	2.2	14	9	1066
conditions $\mathcal{D}$	until unique	100.0%	150	279	0.00	0.0	1	1	–	–	8	17
$\Delta z = 1$ only	72+72+72	100.0%	216	216	2.33	11.8	22	3456	2.0	17	13	358
$\Delta z = 1$ only	until unique	100.0%	291	520	0.00	0.0	1	1	–	–	15	27

9 Robustness of the PND filter (Section III-F)

Table 16: Behaviour of the PND filter under faults outside the model. Harmful: accepted and the derived conditions exclude the true key.

injected fault	injections	accepted	rate	harmful	rate
one PND in z (the model)	20000	20000	100.00%	0	0.00%
two independent PNDs in z	20000	2423	12.12%	4	0.02%
random byte window in z	20000	3634	18.17%	0	0.00%
random 32-bit value in z	20000	0	0.00%	0	0.00%
PND in z and PND in x	20000	2630	13.15%	2630	13.15%
PND in z and PND in w	20000	2578	12.89%	2353	11.77%
PND in z, one round too early	20000	29	0.14%	19	0.10%
PND in z, one round too late	20000	0	0.00%	0	0.00%
$L_2(e_q)$: three spread bits in z	20000	3670	18.35%	3670	18.35%

10 Column decomposition versus 16-bit groups (Section III-C)

Table 17: For each instance, the candidate set of every 16-bit group obtained by exhaustive search over the 16 real key bits equals the product of the four column candidate sets.

seed	PNDs	brute force over 2^{16} equals product of column sets
7000	6	yes
7001	8	yes
7002	10	yes
7003	12	yes
7004	14	yes
7005	16	yes
7006	18	yes
7007	20	yes
7008	6	yes
7009	8	yes
7010	10	yes
7011	12	yes
7012	14	yes
7013	16	yes
7014	18	yes
7015	20	yes
7016	6	yes
7017	8	yes
7018	10	yes
7019	12	yes

11 Regenerating this material

```
python3 run_experiments.py all --trials 10000 --master-trials 1000 --seed 20251101
python3 verify_groups_20.py
python3 make_supplementary.py
cd supplementary && latexmk -pdf supplementary.tex
```